

Since the dawn of time, humans have discovered and innovated. Technology has always been a primary driving force in the advancement of humanity, from the creation of the wheel to artificial intelligence. These developments have innumerable benefits and bring about unprecedented convenience to complex and intensive human tasks. But how far is too far? The industrial growth of factories demonstrated the automation of human “doing”. Artificial intelligence is currently proving the unbelievable potential of how human “thinking” can be automated. It is important to truly understand and consider the consequences of the advancement of technology, before the very essence of what it means to be human and “being” is taken away from us. In his article *More-Than-Human Science*, Brandon Boesch explores this topic and poses the question of whether humans should intentionally devote financial and intellectual resources towards the development of this “automated scenario” powered through artificial superintelligence. This essay will examine both the supporting and opposing arguments of this dilemma and propose a potential resolution.

The tremendous promise of the development of this technology is undeniable. AI can bring immense progress to computationally-intense processes, in science and beyond. Boesch writes how the advantages of computer-driven systems include it being less time-consuming and labor-intensive, and that it is cheaper (Boesch, *Aeon* 2025). In his article, Boesch mentions the revolutionary creation of AlphaFold, an artificial intelligence designed to model and predict proteins and their structure. Traditional protein modeling, like X-ray crystallography can cost up to \$10,000 per protein structure, and doing this process manually is a lengthy process. Boesch writes how even “[t]he creators of AlphaFold—Google DeepMind—claim that it has saved

“hundreds of millions of years in research time” (Boesch, *Aeon* 2025). The reality of scientific research is that it is a slow and intensive process with vast and lengthy data collection and analysis procedures. Having worked on a research project in the computational neuroscience space, I can speak from experience that the data aggregation and pre-processing period takes a high, almost disproportionate amount of time compared to the rest of the scientific procedure. Artificial intelligence and other new technologies can greatly accelerate the pace of scientific research, along with other real world computationally-heavy processes like administrative tasks, accounting, airlines seating, etc. Allocating resources to the development of higher-powered computing can make our world easier and more efficient.

Moreover, humans have a moral responsibility to protect other humans. Increased computing power in the medical healthcare space is an absolute game-changer. More promising than AlphaFold’s saving of “hundreds of millions of years” is the potential to save even more lives with this technology (Boesch, *Aeon* 2025). AlphaFold is already unlocking the key to countless discoveries, and this is just the onset of what’s to come. Self-directed artificial superintelligence can design and run its own research projects to provide progress for humans. Research projects are conducted in small niches too, which means that in order to form generalizable conclusions, studies may need to be replicated amongst thousands of datasets. This technology can launch millions of studies on unique brain tumors, cancers, and other bodily malfunctions and disorders, scraping through hundreds of thousands of datasets instantly—what could have taken hundreds of scientists decades to complete, and can heal thousands of bodily dysfunctions.

However, it is important to consider the reasons opposed to the development of these computing technologies. Offering artificial superintelligence too much independence can easily

lead to unintended consequences. Boesch writes about the negative effects artificial superintelligence could bring like horrific weaponry and potential “human enslavement or annihilation” (Boesch, *Aeon* 2025). Robots taking over Earth may sound like the trope of a science-fiction movie, but could become a reality. For example, in a scenario that computational-powered beings calculate that humans deplete the planet’s natural resources, these machines may decide that human destruction and eradication is mathematically needed to optimize the world. The ethical and philosophical consequences must also be considered with the potential for computationally-powered species to replace humans. What purpose would humans have if society is run by this new quasi-omniscient species? Curiosity is one of humans’ most innate trait, and is manifested the scientific realm as the foundation of all discovery and innovation. If artificial superintelligence takes over science, learning more that is incomprehensible to humans, these scientific endeavors may become disconnected from humans. If something as human as science can be stripped away from us, what next? Technology will have crossed the point of no return and humanity will forever be taken away.

Despite this, however, I believe we must cautiously invest in this new technology because of the inevitability of artificial superintelligence and the automated scenario. Whether we like it or not, the technology is advancing and it is impossible to block advancements completely. Instead of attempting to stop the development of this technology, it is more useful to focus on making this development safe for humans and the world. In 2020, data centers around the world providing needed infrastructure to support AI used approximately 200 tera-watt hours of electricity, contributing to 2% of global greenhouse gas emissions. This number is only expected rise as we expand applications of this new technology. As we experiment with this incredibly powerful technology, we must tread carefully as there are multiple costs at risk with mindless

technological development. I believe the most logical solution worth pursuing is a human-directed or supervised yet AI-powered scientific scenario, with the judgement of humans but the computing power of the automated scenario. This being said, it will be incredibly hard to set boundaries over how much of science should be powered by humans versus computers but is something we must do to push the frontiers of artificial superintelligence slowly and carefully. Society must cross important barriers before fully integrating this advanced computing technology in industries and the real world, including the creation of comprehensive legislation regarding ethical and unethical uses of this technology, transparency and teachings of this technology to the public, safeguards against the stripping of jobs, etc. In Boesch's article, he writes in the conditional tense "if we are to remain human" in the future (Boesch, *Aeon* 2025). The hypothetical nature of this phrasing alludes to something of a doomsday scenario, but I am confident that humanity will persevere, just like we have done for hundreds of thousands of years. I truly believe that the advancements of science, no matter how scary they may seem, pose incredible potential to positively change the world—as long as we ensure the necessary protections against its misuse.